THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 10: **The Other Side of Zero** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 22	Simplify step by step.
2	**(a)** 30	Opposites cancel.
3	**(b)** 23	Distance $= \lvert a - b \rvert$.
4	**(b)** -17	Subtract.
5	**(c)** -30	Gains positive, losses negative.
6	**(c)** -6	The first two cancel.
7	**(c)** -5	Add b to both sides.
8	**(d)** 2	Subtracting a negative adds.
9	**(a)** Both A and R are true and R is the correct explanation of A.	-7 lies to the left of -3.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true, but R does not explain why 5 + (-5) = 0.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Opposite integers cancel each other; Remaining: integers from 7 to 10 (1); Sum $= 34$ (1).	1+1
12	$? = 5 - (-9)$ (1); $? = 14$ (1).	1+1
13	Final: $-20 + 60 - 25 = 15$ m (1); Distance from 30 m: 15 m (1).	1+1
14	Opposite integers cancel each other; Remaining: integers from 10 to 14 (1); Sum $= 60$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	After dive: -80; After rise: -40 (1); After second dive: -65 (1); Total distance $= 115$ m (1).	1+1+1
16	Other $= -4 - 11 = -15$ (1); Difference $= 26$ (1); Check the sum: -4 (1).	1+1+1
17	Common difference $= 6$ (1); Next terms: 4, 10, 16 (1); First positive term: 4 (1).	1+1+1
	OR: After dive: -80; After rise: -25 (1); After second dive: -55 (1); Total distance $= 155$ m (1).	1+1+1
18	Other $= 9 - (-6) = 15$ (1); Difference $= 21$ (1); Check the sum: 9 (1).	1+1+1
	OR: Common difference $= 4$ (1); Next terms: 1, 5, 9 (1); First positive term: 1 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Tuesday ($-5°$C) (1). (ii) $1 - (-5) = 6°$C (1). (iii) Monday, by $-2 - (-3) = 1°$C (1). (iv) $-2 + (-5) + 1 + (-3) = -9$ (2).	1+1+1+2
20	Scores: -4, -11, 4, -2 (1); Final score -2 (1); Lowest score -11 (1); Write all values as integers (2).	1+1+1+2
	OR: After down: -2 (1); After up: -1 (1); After down: -3 (1); Does not exist (below the lowest basement) (2).	1+1+1+2

Mistakes to avoid:

- Saying $-7 > -3$ because 7 is greater than 3; on the number line -7 lies to the left of -3.
- Writing $-4 - (-6) = -10$; the correct value is $-4 + 6 = 2$.
- Dropping the sign of the bigger number when adding integers with different signs, e.g. $(-9) + 5 = 4$ instead of -4.
- Taking the successor of a negative integer as one smaller (successor of -1 is 0, not -2).
- Finding a fall or rise as initial − final in the wrong order; use final − initial and read the sign.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 2

Chapter 10: **The Other Side of Zero** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 42	Opposites cancel.
2	**(d)** 16	Distance $= \lvert a - b \rvert$.
3	**(b)** 17	Subtract.
4	**(b)** -110	Gains positive, losses negative.
5	**(d)** -9	The first two cancel.
6	**(c)** 11	Add b to both sides.
7	**(d)** -22	Subtracting a negative adds.
8	**(c)** 10	All pairs cancel except the last.
9	**(c)** A is true but R is false.	-4 + 6 = 2 is true. Subtracting -3 is the same as adding 3, so R is false.
10	**(d)** A is false but R is true.	-1 + 1 = 0, so the successor of -1 is 0 and A is false. R is true.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$? = -6 - 14$ (1); $? = -20$ (1).	1+1
12	Final: $-45 + 100 - 80 = -25$ m (1); Distance from -10 m: 15 m (1).	1+1
13	Opposite integers cancel each other; Remaining: integers from 13 to 15 (1); Sum $= 42$ (1).	1+1
14	$? = -8 - (-21)$ (1); $? = 13$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	After dive: -70; After rise: -10 (1); After second dive: -25 (1); Total distance $= 100$ m (1).	1+1+1
16	Other $= -15 - (-7) = -8$ (1); Difference $= 1$ (1); Check the sum: -15 (1).	1+1+1
17	Common difference $= 7$ (1); Next terms: -2, 5, 12 (1); First positive term: 5 (1).	1+1+1
	OR: After dive: -115; After rise: -95 (1); After second dive: -140 (1); Total distance $= 100$ m (1).	1+1+1
18	Other $= 20 - 35 = -15$ (1); Difference $= 50$ (1); Check the sum: 20 (1).	1+1+1
	OR: Common difference $= 5$ (1); Next terms: 2, 7, 12 (1); First positive term: 2 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) -3 (1). (ii) $-3 + 5 = 2$ (1). (iii) $2 - 4 = -2$ (1). (iv) Total travel $= 3 + 5 + 4 = 12$ floors (1); it finishes at -2, the second basement (1).	1+1+1+2
20	Scores: 5, -5, -1, -6 (1); Final score -6 (1); Lowest score -6 (1); Write all values as integers (2).	1+1+1+2
	OR: After down: -2 (1); After up: -1 (1); After down: -5 (1); Does not exist (below the lowest basement) (2).	1+1+1+2

Mistakes to avoid:

- Saying $-7 > -3$ because 7 is greater than 3; on the number line -7 lies to the left of -3.
- Writing $-4 - (-6) = -10$; the correct value is $-4 + 6 = 2$.
- Dropping the sign of the bigger number when adding integers with different signs, e.g. $(-9) + 5 = 4$ instead of -4.
- Taking the successor of a negative integer as one smaller (successor of -1 is 0, not -2).
- Finding a fall or rise as initial − final in the wrong order; use final − initial and read the sign.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 3

Chapter 10: **The Other Side of Zero** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 23	Distance $= \lvert a - b \rvert$.
2	**(a)** -14	Subtract.
3	**(d)** -250	Gains positive, losses negative.
4	**(d)** -12	The first two cancel.
5	**(c)** 15	Add b to both sides.
6	**(c)** 36	Subtracting a negative adds.
7	**(a)** 21	Simplify step by step.
8	**(c)** 92	Opposites cancel.
9	**(a)** Both A and R are true and R is the correct explanation of A.	The difference between the readings gives the fall.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true, but R does not explain why the sum is negative.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Final: $-15 + 40 - 60 = -35$ m (1); Distance from 20 m: 55 m (1).	1+1
12	Opposite integers cancel each other; Remaining: integers from 9 to 13 (1); Sum $= 55$ (1).	1+1
13	$? = 3 - 17$ (1); $? = -14$ (1).	1+1
14	Final: $-30 + 90 - 50 = 10$ m (1); Distance from 25 m: 15 m (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	After dive: -90; After rise: -60 (1); After second dive: -100 (1); Total distance $= 135$ m (1).	1+1+1
16	Other $= -8 - 13 = -21$ (1); Difference $= 34$ (1); Check the sum: -8 (1).	1+1+1
17	Common difference $= 8$ (1); Next terms: 7, 15, 23 (1); First positive term: 7 (1).	1+1+1
	OR: After dive: -105; After rise: -30 (1); After second dive: -50 (1); Total distance $= 140$ m (1).	1+1+1
18	Other $= 2 - (-18) = 20$ (1); Difference $= 38$ (1); Check the sum: 2 (1).	1+1+1
	OR: Common difference $= 3$ (1); Next terms: 0, 3, 6 (1); First positive term: 3 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Tuesday and Wednesday (1). (ii) $-120 + (-80) = -200$, a loss of ₹200 (1). (iii) $250 - 120 - 80 + 300 = 350$, a profit of ₹350 (1). (iv) $350 + x = 0$ gives $x = -350$ (1); a loss of ₹350 (1).	1+1+1+2
20	Scores: -5, 6, -7, -2 (1); Final score -2 (1); Lowest score -7 (1); Write all values as integers (2).	1+1+1+2
	OR: After down: -2 (1); After up: 0 (1); After down: -3 (1); Does not exist (below the lowest basement) (2).	1+1+1+2

Mistakes to avoid:

- Saying $-7 > -3$ because 7 is greater than 3; on the number line -7 lies to the left of -3.
- Writing $-4 - (-6) = -10$; the correct value is $-4 + 6 = 2$.
- Dropping the sign of the bigger number when adding integers with different signs, e.g. $(-9) + 5 = 4$ instead of -4.
- Taking the successor of a negative integer as one smaller (successor of -1 is 0, not -2).
- Finding a fall or rise as initial − final in the wrong order; use final − initial and read the sign.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.